

Multi-Tenant SaaS POS System

**A SYNOPSIS OF MAJOR PROJECT
for the Degree of**

**BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE & ENGINEERING
SUBMITTED TO**



**LNCT UNIVERSITY
Bhopal (M.P.)**

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Year - 3rd
semester - 6th
Session: JAN-JUNE 2026

Synopsis

Project Title

Multi-Tenant SaaS POS System

1.Problem Statement

Traditional Point of Sale (POS) systems used by small and large retail businesses are often expensive, hardware-dependent, difficult to scale, and lack centralized management. Many existing solutions do not support multi-branch operations, real-time inventory tracking, role-based dashboards, or modern payment integrations.

Retail chains, supermarkets, and malls require a cloud-based, scalable, secure, and multi-tenant POS system that can manage sales, inventory, employees, customers, and analytics efficiently from a single platform.

2. Objective of the Project

The primary objective of this project is to design and develop a production-ready enterprise SaaS POS system that automates retail operations using modern full-stack technologies.

Specific Objectives:

- To develop a multi-tenant POS platform for multiple stores and branches
- To provide role-based dashboards for Cashier, Branch Manager, and Store Admin
- To manage sales, inventory, refunds, customers, and employees
- To implement secure authentication and authorization
- To support multiple payment gateways
- To generate real-time reports and analytics
- To create a scalable SaaS subscription model

3. Proposed Solution

The proposed solution is a web-based SaaS Point of Sale system built using React for frontend, Spring Boot for backend, and MySQL as the database.

The system follows a Client–Server Architecture where:

- The frontend provides interactive dashboards and cashier interfaces
- The backend handles business logic, security, and API processing
- The database stores transactional and master data securely

The solution supports:

- Multi-branch operations
- Centralized inventory tracking
- Secure payment processing
- Business intelligence dashboards
- Subscription-based SaaS model

4. Scope of the Project

- Retail shops, supermarkets, malls, franchises
- Multi-branch and multi-user support
- Inventory and stock management
- Sales, refunds, and shift reports
- Role-based dashboards
- Payment integration
- Scalable SaaS architecture

5. System Requirements

Hardware Requirements

- Processor: Intel i5 or higher
- RAM: Minimum 8 GB
- Storage: 256 GB SSD or higher
- Internet Connection: Required

Software Requirements

- Operating System: Windows / Linux / macOS
- Frontend Runtime: Node.js (v18+)
- Backend Runtime: Java JDK 17+
- Database Server: MySQL 8+
- IDE: VS Code / IntelliJ IDEA
- Browser: Google Chrome / Firefox

6. Technology Stack

<u>Layer</u>	<u>Technology</u>
Frontend	React.js, Redux Toolkit (16 reducers), Axios ,shadcn/ui, Tailwind CSS
Backend	Spring Boot, Spring Security, JPA / Hibernate, RESTful APIs
Database	MongoDB
API	RESTful Web Services
Tools	·Maven ,Git & GitHub, Postman , Cloudinary (image storage), Razorpay & Stripe (payments)

7. System Features

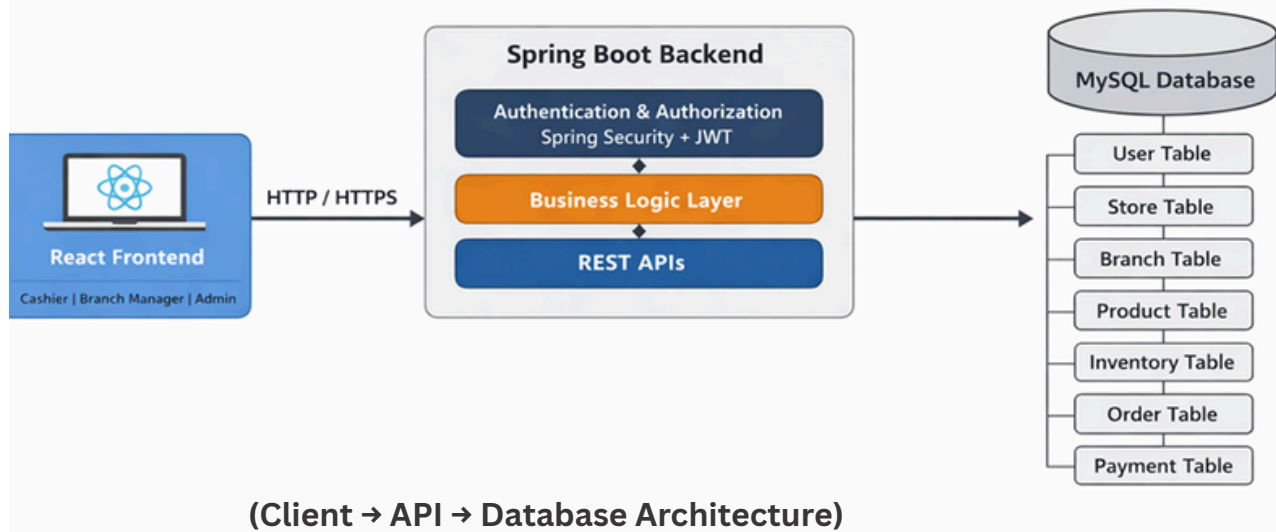
Cashier Terminal Interface • Product search with barcode scanning • Smart shopping cart with quantity management • Multiple payment methods (Card, Cash, UPI) • Customer selection & management • Discount system (percentage or fixed amount) • Order notes & special instructions • Refund processing • Complete order history with date filters • Invoice generation & printing • Shift reports with payment breakdown	Store Admin Panel • Multi-location management (unlimited stores & branches) • Enterprise-wide analytics dashboard • Employee management across all locations • Product catalog with categories • Customer loyalty points system • Sales trends & growth tracking • Subscription plan management • KPI tracking (Total sales, Average order value, etc.)
Branch Manager Dashboard • Order management with search & filters • Real-time inventory tracking • Low stock alerts with product details • Employee management (active/inactive status) • Branch analytics & performance metrics • Top selling products tracking • Branch settings & configuration • Payment method breakdown • Intelligent busin • ess alerts	Business Intelligence Features • Daily sales trends (Last 7 days) • Payment method distribution charts • Branch performance comparison • Customer purchase history • Sales growth percentage tracking • Average order value analytics • Top products & categories reports

8. Proposed Methodology

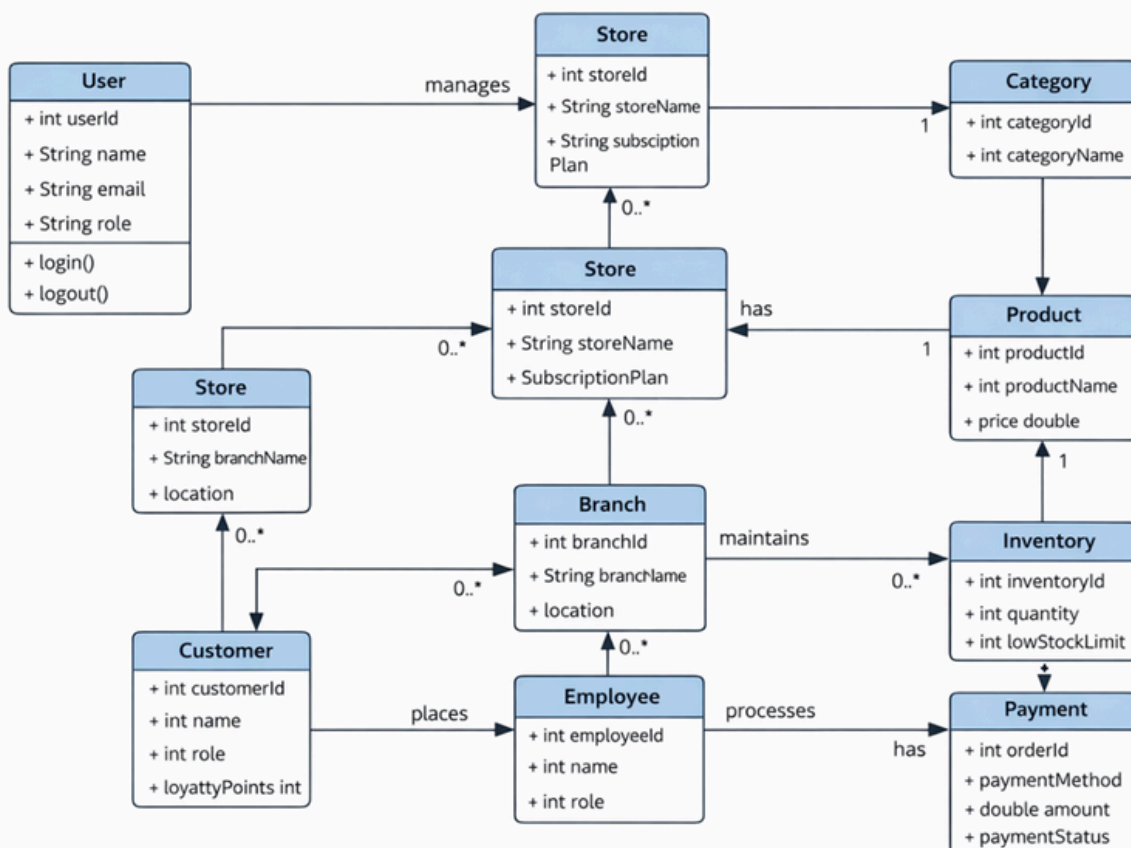
The project follows a modular and incremental development methodology:

1. Requirement Analysis
2. System Architecture Design
3. Database Design (ER Diagram)
4. Backend API Development
5. Authentication & Security Implementation
6. Frontend UI Development
7. Redux State Integration
8. Payment Gateway Integration
9. System Testing
10. Deployment Preparation

- System Architecture Diagram



- ER Diagram / Database Design



9. Expected Outcome

- Fully functional enterprise SaaS POS system
- Secure and scalable architecture
- Real-time inventory and sales tracking
- Professional dashboards for different user roles
- Portfolio-ready production-grade project
- Improved understanding of enterprise application development

10. Future Scope

- Mobile POS application (Android/iOS)
- Barcode & QR code scanner integration
- AI-based sales forecasting
- GST and tax automation
- Cloud deployment (AWS/GCP)
- Multi-language support
- Offline mode support